

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

Pentester Name	Mcebisi Nsele
Role	Cybersecurity Professional (Intern)
Program/Batch	B083E
Date	16/09/2026
Modules Completed	PM1 (Kali Linux Footprinting Tools), PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (networkwalks.com) — secured written permission already 2. My own local LAN network
Permission Secured From Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using six Kali Linux tools (WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon) and scanning my own local network with Zenmap. One activity covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is Week 2 of my ongoing internship in cybersecurity.

All footprinting commands were run in Kali Linux. The network scan was run using Zenmap (Nmap GUI) on windows 11. Every step below includes the exact command/action used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).

WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP).
Nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
Wafw00f	Detect whether a Web Application Firewall protects the site.
DNSRecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
ipconfig / Local CMD	Identify local IP address and subnet before scanning.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information and identify the domain's name servers. The results showed the domain is registered with GoDaddy.com, LLC, created on 6 November 2019 and set to expire on 6 November 2027. The registrant details are hidden behind Domains By Proxy, LLC privacy protection, and the domain uses ns6135.hostgator.com and ns6136.hostgator.com (plus ns29/ns30.domaincontrol.com) as name servers. DNSSEC is unsigned.

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.1 as the CMS, along with the WordPress Download Manager 3.3.58 plugin, Bootstrap 7.1, jQuery 3.7.1, an Apache server, Google Tag Manager, and a publicly listed contact address (info@networkwalks.com). The site's IP address, 192.232.216.135, and its hosting country (United States) were also exposed. The HTTP request to the site further showed a 301 redirect from http:// to the https:// version.

Using Nslookup, I resolved the domain name to its IP address via Google's public DNS resolver (8.8.8.8). The result identified 192.232.216.135, matching the IP found by WhatWeb.

I used Curl with the -I option to inspect the HTTP response headers. The response returned HTTP/2 200, an Apache server banner, and a Link header exposing the WordPress REST API endpoints (/wp-json/ and /wp-json/wp/v2/pages/53). It also set an HttpOnly, secure session cookie (__wpdm_client) and included a permissions-policy header plus caching headers (x-nginx-cache: WordPress) indicating the site runs behind Nginx-based caching in front of Apache.

Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs) WAF protecting the site.

Finally, I used DNSRecon to enumerate DNS records. The results identified the SOA and NS records (ns6135.hostgator.com / 50.87.144.87 and ns6136.hostgator.com / 192.232.216.131), an MX record (mail.networkwalks.com), an A record confirming 192.232.216.135, a Google site-verification TXT record, an SPF record (v=spf1 +a +mx +ip4:50.87.144.87 +include: websitewelcome.com ~all), and eight SRV records for Autodiscover pointing to cpanelmaildiscovery.cpanel.net. DNSRecon also disclosed the BIND version (9.16.23-RH) running on both name servers.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap to perform network discovery on my own local network. This required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the local ipconfig command to identify my local IP address and LAN subnet: **172.30.96.1 & 255.255.240.0**, I then entered the subnet into Zenmap and selected a Ping Scan to identify active hosts.

The scan identified the following live hosts on my local network:

- **172.30.96.1** — MAC address - **00-15-5D-09-BD-80**

After completing the scan, I opened the Topology section in Zenmap, enabled the legend, and saved the network topology in PDF format.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1 and WP Download Manager 3.3.58, plus Bootstrap and jQuery versions	Attackers may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup and WhatWeb both resolved the domain to 192.232.216.135	Provides information about the network location of the web service	Low
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed the WordPress REST API endpoint (/wp-json/)	May assist technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs) protecting the site	Reveals information about the web application's security architecture	Low
5	DNS & mail infrastructure information exposed	DNSRecon identified NS, MX, SPF/TXT and SRV (Autodiscover) records	DNS information can help build a broader infrastructure profile	Medium
6	DNS server software version disclosed	DNSRecon identified BIND version 9.16.23-RH on both name servers	Version-specific information could be used to check for known DNS server vulnerabilities	Medium

7	Multiple live hosts visible on local network	Zenmap identified [number] live hosts on my local network	Unknown or unauthorized devices may potentially be present on a network	Medium
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Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two activities.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from this activity, I recommend the following security improvements:

- Review publicly exposed technology information — organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
- Keep software updated — CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- Review HTTP headers — response headers should be checked to determine whether unnecessary technical information is being exposed.
- Review DNS records regularly — DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
- Suppress DNS server version banners — configure name servers to hide or restrict version.bind responses so software versions aren't easily enumerable.
- Properly configure and monitor the WAF — keep the WAF enabled and tuned so it continues to block naive attacks.
- Perform regular internal network discovery — periodically scan local networks to identify active devices.
- Investigate unknown devices — any unexpected device discovered during network scanning should be investigated and verified.
- Maintain network documentation — network topology and device information should be documented and updated regularly.
- Perform security testing with authorization — reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies,

Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

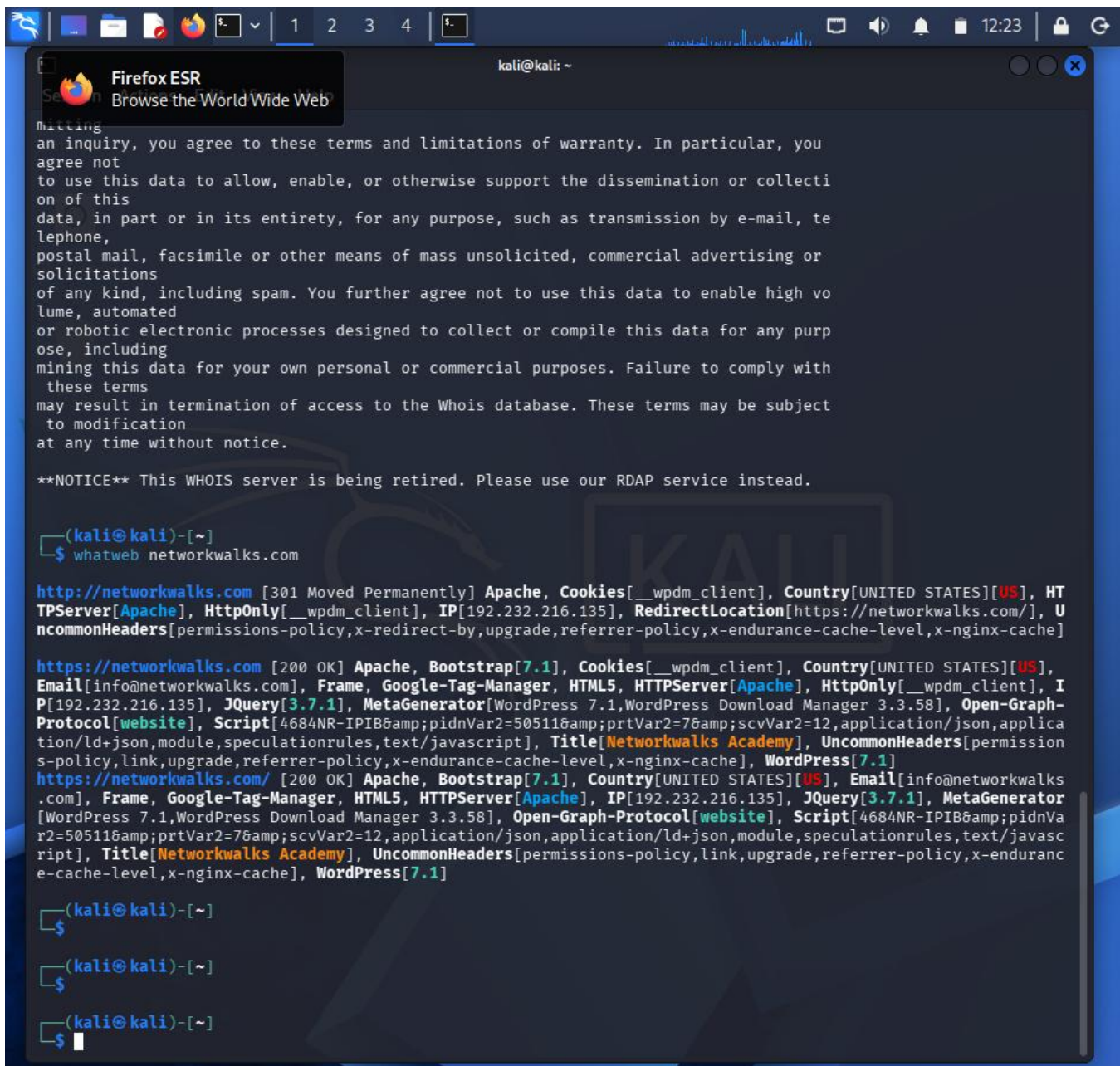
In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

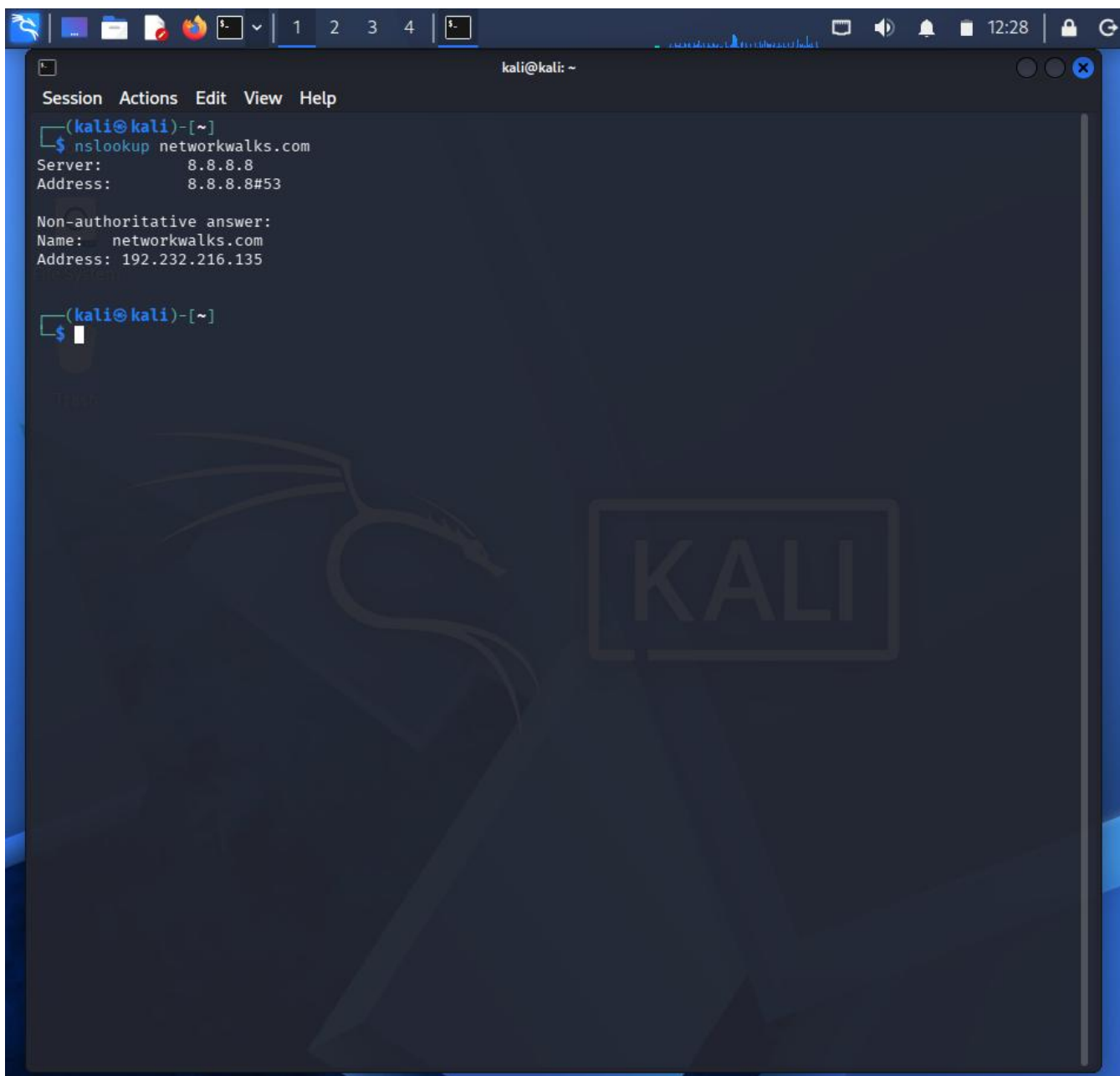
The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to assess a system further, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

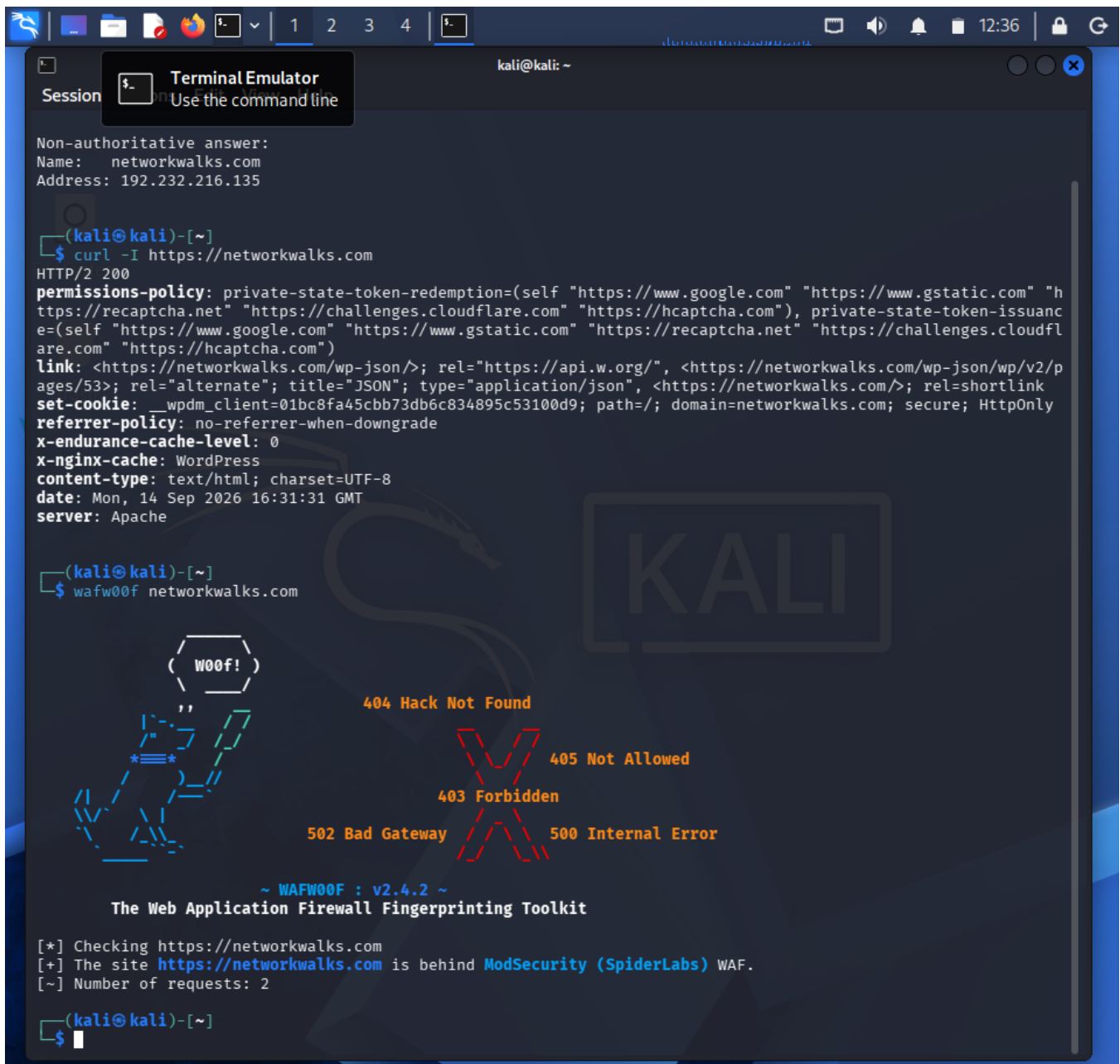
I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of an assigned educational cybersecurity lab.

8. Evidences Collected

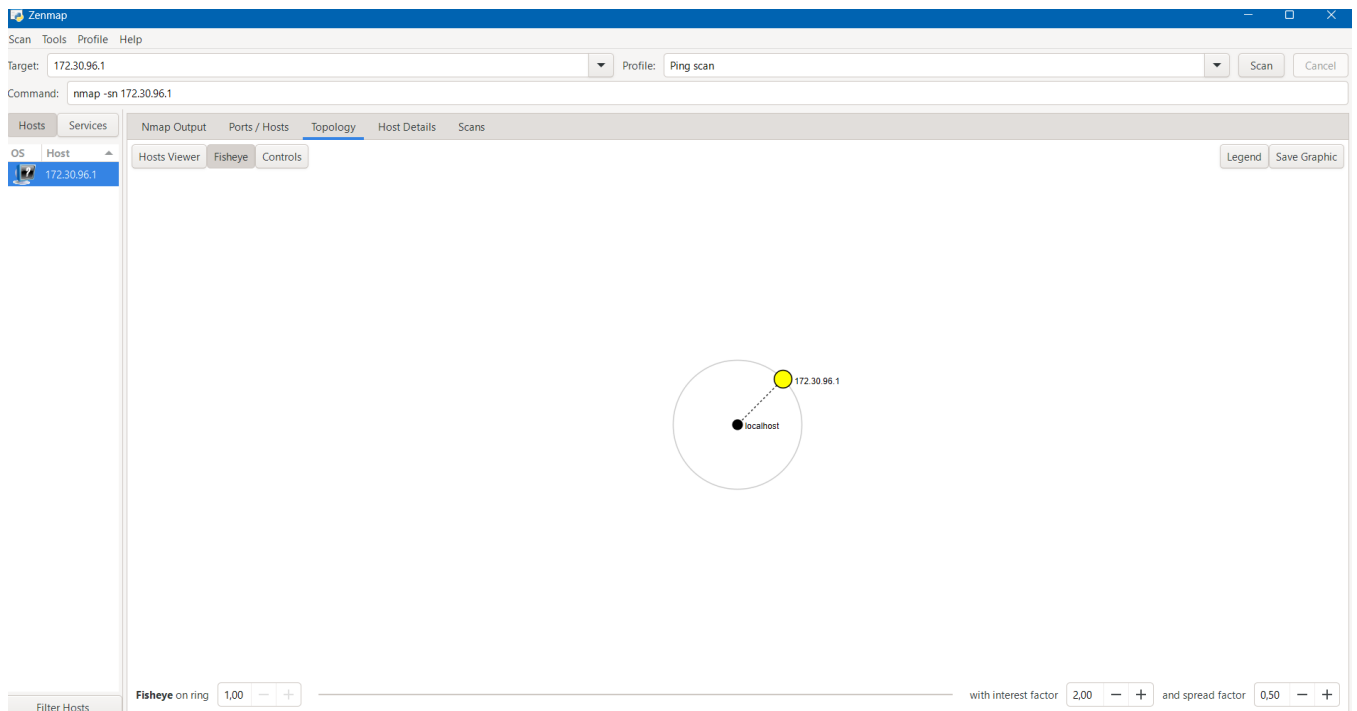






```
kali@kali: ~  
Session [x] Terminal Emulator  
Use the command line  
(kali@kali)-[~]  
$ dnsrecon -d networkwalks.com  
2026-09-14T12:38:56.288113-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-09-14T12:38:56.291917-0400 INFO std: Performing General Enumeration against: networkwalks.com...  
2026-09-14T12:38:57.006869-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-09-14T12:38:58.021911-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-09-14T12:38:58.660905-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-09-14T12:38:59.483680-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-09-14T12:38:59.484852-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-09-14T12:39:00.285399-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-09-14T12:39:01.407914-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-09-14T12:39:01.616927-0400 INFO A networkwalks.com 192.232.216.135  
2026-09-14T12:39:02.534236-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK  
4q75X-Ij-mjgEeg-UsYI  
2026-09-14T12:39:02.534647-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:website  
welcome.com ~all  
2026-09-14T12:39:03.462225-0400 INFO Enumerating SRV Records  
2026-09-14T12:39:10.849695-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.203.15 443  
2026-09-14T12:39:10.850382-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.204.9 443  
2026-09-14T12:39:10.850687-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.203.11 443  
2026-09-14T12:39:10.850860-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.203.9 443  
2026-09-14T12:39:10.851013-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.204.15 443  
2026-09-14T12:39:10.851270-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.204.11 443  
2026-09-14T12:39:10.851420-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.204.14 443  
2026-09-14T12:39:10.851970-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.n  
et 184.94.203.14 443  
2026-09-14T12:39:10.855490-0400 INFO 8 Records Found  
2026-09-14T12:39:10.856029-0400 INFO Completed enumeration for domain: networkwalks.com  
(kali@kali)-[~]  
$
```

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-14T16:13:37Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to
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Zenmap

Scan Tools Profile Help

Target: 172.30.96.1 Profile: Ping scan Scan Cancel

Command: nmap -sn 172.30.96.1

Hosts Services

OS Host 172.30.96.1

Nmap Output Ports / Hosts Topology Host Details Scans

Hosts Viewer Fisheye Controls Legend Save Graphic

Filter Hosts

172.30.96.1

Starting Nmap 7.991 (<https://nmap.org>) at 2026-09-16 22:32 +0200
Nmap scan report for 172.30.96.1
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 1.08 seconds